The XRMB (on MR450 and MR750) and XRMw (on "–w" systems) gradient coils converts the X, Y, and Z gradient currents into gradient fields.

The PDU (Power Distribution Unit) provides a multi-tap 3ϕ transformer to convert a variety of line voltages to the 420V and 208V needed by the system. It provides isolation between the site power and the system, and provides breakers for distributing power throughout the system.

Section 3.4 covers the basic theory of the gradient field. Section 3.7 expands further to cover the gradient subsystem. Section 3.7.2 covers the basic theory of the gradient voltage and current waveforms.

4.6.1 Gradient subsystem performance parameters

High level gradient subsystem performance parameters for recent GEHC MR scanners are tabulated below.

Product	*G*, mT/m	*SR*, T/m/s	Bore, cm	Coil	Driver
MR450/MR750	50	200	60	XRMB	XGD
MR450w/MR750w	33	120	70	XRMW	XG2
Rio (initial release)	70	150	70	HRMW2	RGD

MR750 and MR450 – the 60cm systems – use the same gradient platform: XGD drivers and XRMB coils. Similarly, the "widebore" or 70cm systems – MR450w and MR750w – use the XG2 drivers, but with XRMw coils. Therefore, high level gradient subsystem performance parameters follow the 60cm vs. 70cm configurations. Note that the unlike the appearance, 60cm systems are not explicitly associated with the XGD driver. The XG2 driver replaced the XGD driver. After release, ALL DV systems are shipped with the XG2 driver. Rio was designed to combine 60cm gradient performance with a 70cm bore configuration.

4.6.1.1 Config parameters

The following parameters (among many others) are store in `GradientConfig.cfg`:

Parameter	*MR450 MR750*	*MR450w MR750w*	*Rio*	*Description*
`<xyz>FSAmp`	`5.0`	`3.3`	`7.0`	The full scale gradient strength, in G/cm
`<xyz>RiseTime`	`248`	`276`	`468`	The time for a rising gradient ramp, in µs
`<xyz>FallTime`	`248`	`276`	`468`	The time for a falling gradient ramp, in µs
`<xyz>Full`	`~30250`	`~22300`	`~24500`	The 16bit value needed for full scale gradient
`coilAC_gain`	`124`	`194`	`129`	Gradient coil gain, in Amps per G/cm

These parameters set the *default characteristics* for gradient waveforms used by the system. For a system configured with `xFSAmp` = 5.0, all gradient pulses will be limited to 5.0 G/cm, even if the gradient subsystem may be capable of higher performance.

4.6.1.2 Flowdown

The gradient driver and gradient coil work together to meet the subsystem specifications. However, they are somewhat independent designs, so understanding how subsystem specification flow down to component specs is useful.

GES flowdown

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In the most basic sense, *GFS* is the gradient coil gain K_{GC} (in G/cm/A) multiplied by the gradient driver's full scale current *IFS* (in A). Note that "full scale current" is the current needed to reach *GFS*, as opposed to the maximum output current for a gradient driver. In addition, there are other factors that need to be considered in determining the subsystem's gradient strength: in significance order, these are eddy current demands and grad shim current demands. For systems where there is the maximum output current is significantly higher than *IFS*, these demands can be covered by this difference. However, the design must to ensure that these demands are accounted for in the gradient current budget.

SR flowdown

Slew rate flowdown is much more complex. Slew rates are limited by two effects. First, the maximum driver voltage must overcome the coil's inductance; this $L\,di/dt$ term limits slew rate.

4.6.1.3 Capability models

The gradient subsystem's capability is key to the overall system's imaging capability. The subsystem's capability considers many varied factors; power semiconductor temperature, gradient coil power dissipation, and AC power draw from the power line are among them. Since the gradient subsystem accounts for a large part of ICV, installation, and operating expenses, accurately predicting and managing the subsystem capability is critical to system performance.

4.6.1.3.1 DV Capability Models

4.6.1.3.2 Rio Capability Models

The Rio gradient subsystem capability models are the most complete models created, as of mid-2017. They incorporate limits on gradient coil power deposition, semiconductor temperature, AC power draw, cooling capabilities, etc. Below is a high-level summary of the capability model implementation and assumptions used for Rio. Note these do not include patient safety related limits such as SAR and dBdt. Refer to DOC1586704 for more details. Refer to DOC2029395 for the list of model parameter values used when setting up pulse sequence protocols. Finally, refer to DOC1691620 for a summary of the impact of these models on pulse sequence performance.

Eddy Current Compensation: Unlike DV, Rio applies a constant 'worst-case' set of short time constant eddy current correction parameters before estimating the waveform power.

RMS Current: The average (root-mean-square) current is capped to protect the amplifiers and cables.

Driver Load: The gradient coil impedance as a function of frequency is modelled using a 10-element ladder model. In addition, the resistance of the longest set of gradient cables and the filter is accounted for.

Driver IGBT junction temperature: The power generated by the IGBTs in the gradient driver is limited based on the semiconductor loss curves.

RC Resistor: The power through a resistor in the RGA output filter from output leads to ground is limited to reduce the chance of field failures.

RPS Power: The peak and average gradient output power is limited assuming worst-case DC coil resistance. The scan is prevented from running if peak is estimated to be over the limit. Dead time is added to the sequence if the average power is estimated to be over the limit.

PDU Power: To estimate the PDU power, the RPS output power plus the losses in the RPS for each axis are added together. The losses within the RPS are a function of the output power. The low line efficiency is used in the model because the output of this model is used to calculate breaker current and breaker current is maximized at low line.

RPS Transformer Current: This model protects the RPS transformer from overheating and from premature aging due to temperature. It estimates the input DC link current based on RPS input power and PDU output voltage. The DC link current is then converted to transformer current.

Chiller Capacity: The model estimates the heat lost to water for the entire system. If the heat lost to water exceeds the cooling circuit capacity, deadtime will be added to reduce the average loss.

4.6.1.4 Gradshim

Since the x, y, and z gradient fields vary linearly with position, they can be used as the x, y, and z first-order shims. This use is called gradient shimming, or just "gradshim", and is generally the only method of first-order shimming available. Gradshims are implemented by adding an offset or DC current onto gradient current waveforms. Changing the offset current changes the amount of gradshim field.

Unlike passive or superconductive shims, gradshims provides real time control. Gradshims are set by the "autoshim" pass of auto-prescan, and user controls are provided on the manual prescan page.

The offset current used by gradshimming is small compared to the full scale gradient current, on the order of 1%.

4.6.1.4.1 Gradshim resolution

Because gradshim currents tend to be small compared to the full scale currents, gradshim's resolution can be a problem.

Shim bit "resolution" refers to the smallest direct-current field gradient change that can be demanded using the imaging gradient windings (X, Y or Z). In other words, it is the field gradient change associated with one count of digitally controlled linear shim. Shim bit resolution can limit the achievable field homogeneity critical for applications such as spectroscopy.

The shim bit resolution is dictated by a) the gain of the gradient winding, b) the maximum gradient driver current and c) the resolution of the gradient current control. A smaller the shim bit resolution implies greater capability (higher resolution) to achieve the desired field gradient.

Shim bit resolution may be calculated using the maximum rated gradient amplitude *GFS* divided by the corresponding digital value that corresponds to GFS. This digital value (set for each axis by *gradcal*) is given by the config file values *xfull*, *yfull*, or *zfull* -- let $cfxyzfull_{NOM}$ be the nominal value:

$$resolution = \frac{GFS}{cfxyzfull_{NOM}}$$

Units are Gauss/cm/bit.

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The current capability target reference is the HDx/BRM system. The Rio system has 2.5X less resolution than HDx/BRM. A summary table:

Platform	*GFS* G/cm	*cfxyzfull* (Nominal)	Resolution G/cm/bit
HDx/BRM	3.3	30,000	110E-6
Rio	7.0	24,500	290E-6

This Rio configuration resulted in a minimum field for the LVShim tool in Gradient Shim mode (22cm DSV) of approximately 13Hz. This prevented several ways from meeting the specifications set in the LVShim tool (maximum absolute field less than 6 Hz).

To achieve the target shim bit resolution, the digital word length used for gradshim control was extended from 16 to 20 bits (as was done in the Firefly program). This change reduces the step size by 16, with the resultant Rio shim bit resolution is 18 uG/cm/bit. This corresponds to a step size for the LVShim tool in Gradient Shim mode (22cm DSV) of less than 1Hz.

Note that Rio still uses 16 bit scaling for the *<xyz>full* values.

4.6.2 Gradient drivers

Gradient drivers accept 3-axis gradient waveform commands from the sequence controller, and translate that into gradient current waveforms.

MR750 and MR750w initially used the XGD gradient driver. XGD can produce up to 660A and 1650V into the XRM coils. The XGD was replaced by the XG2 gradient driver. XG2 was redesigned to simplify (shorten) the signal paths and increase the maximum digital current capability (up to 900A into the XRM coils) at the expense of bit resolution. As a result, the <xyz>full calibration values reduced by a factor of 660/900 and the rfdelay and gradient delay values were reduced. Note the capability of the amplifiers did not change, just the controller design.

Rio introduced the RGD (Rio GD), which was one of the major technology development efforts for Rio. For the Rio Gradient Amplifier (RGA), refer to DOC1747702 for design details and performance characterization result references. The RGA can produce up to 1040 A and 2460V (615V per bus) into the HRM coil). The RGA has 4 identical buses and eight gate drivers (2 per bus). It uses Si IGBTs. It is a switching power supply that operates at 20kHz normally and 10kHz for long gradients. For the details and performance characterization results of the Rio gradient controls, refer to DOC1737385. Rio leveraged the existing (XG2) control board changing analog scaling of voltage and current feedback. Since the busses are at the same voltage, only one controller is used. The controller uses the AC4 design which uses an FPGA (rather than DSP).

4.6.3 Gradient coils

A gradient coil (GC) convert gradient currents into gradient fields.

The premium products use the following gradient coils:

Product	*Name*
MR750, MR450	XRMB

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MR450w, MR750w	XRMW
Rio	HRMW

These are cylindrical, self shielded, single FOV, water cooled gradient coils. The main difference is inner diameters; the "W" coils have larger inner diameter to allow a 70cm patient bore. In addition, XRMB can (as an option) support high order shim coils, HRMW contains high order shim coils and XRMW does not.

GC design must trade off several important parameters, such as gain, FOV, impedance, power handling, and shielding (among others). The design is also constrained physically: outer diameter is limited by magnet inner bore, inner diameter is limited by patient bore size and the space needed for the RF body coil, and the length is limited by patient bore length and overall magnet length. As a result, a GC design is a highly specialized effort, and it can provide major competitive advantage over competing MR systems.

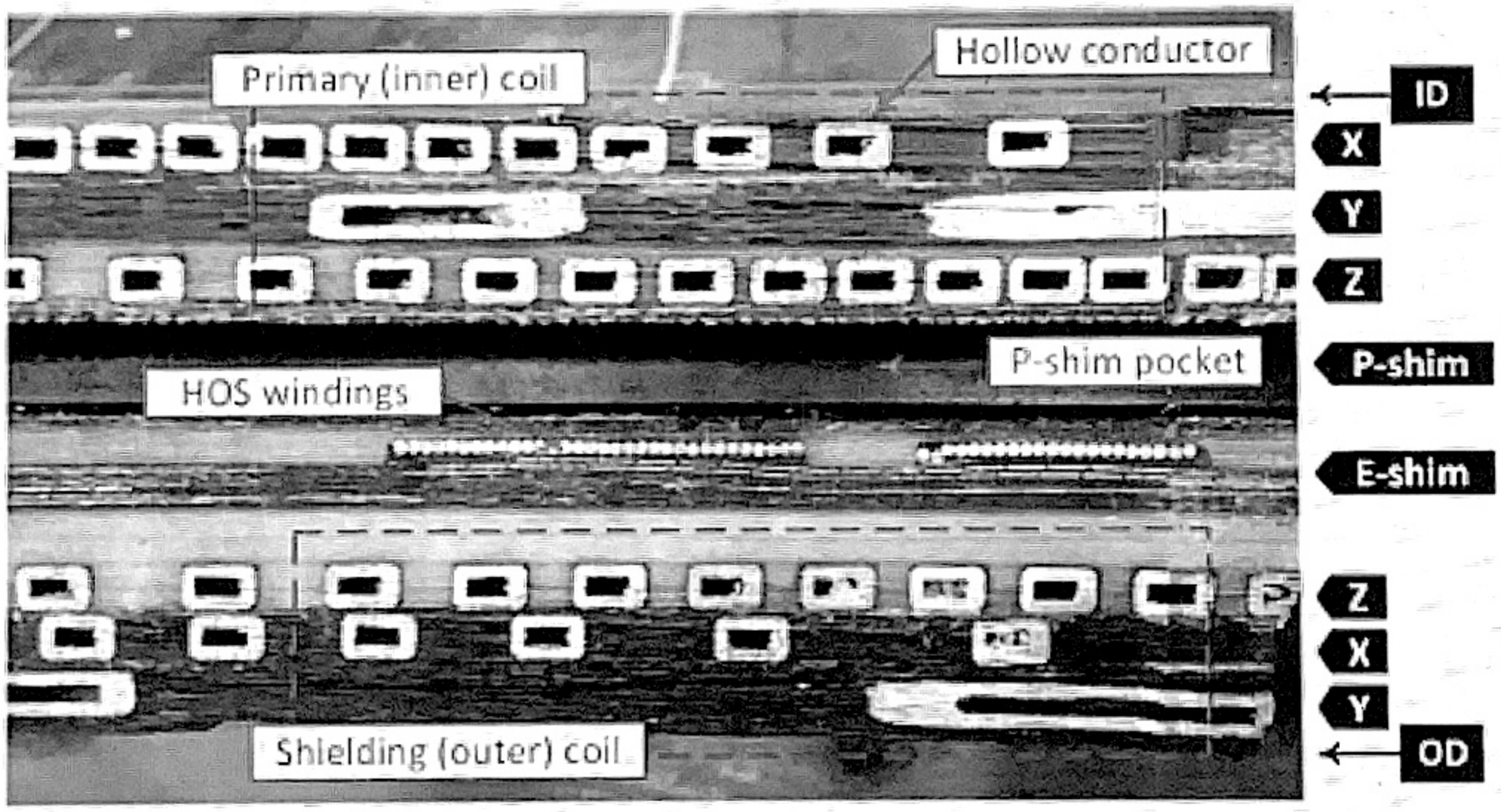


Fig. 4–19: Section of Rio HRMw gradient coil

Gradient coils have many functions, engineered into extremely limited space. Generally, these functions are stacked into layers. The figure above shows a cross-section of an HRMW gradient coil – a radial cut thru the wall. The overall gradient coil is a cylindrical tube, roughly 91cm outer diameter, 74cm inner diameter, and 167cm long; this section is about 30cm long. The inner diameter of the coil is at the top of the photo, and the outer diameter at the bottom.

The gradient windings' conductors can be seen in each of the six windings; Rio's hollow conductors carry coolant when in use. The primary (inner) coil at the top and the outer at the bottom; note that the winding density is much higher for the primary winding than for the secondary. Note that the tubes for the y-gradient appear elongated, since they are cut at a shallow angle.

The "pockets" for holding the passive shim carriers are visible. These pockets are long tubes, running end-to-end on the coil, so they appear as a channel running left-to-right.

Two types of high-order shim windings can be seen, used for real time shim (as called HOS, or E-shim). The arrow points out a wire-wound shim coil, and a printed-circuit winding is directly below it.

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4.6.3.1 Gain

GC gain specifies the gradient field produced per amp of gradient current. A GC with gain of 0.01 G/cm/A would require 100A of current to produce 1 G/cm gradient field.

GC gain is a tradeoff between the current and voltage demands on the driver. A high gain GC requires less current to create a given gradient field. However, higher gain GCs require more turns in their windings, which lead to higher inductance and resistance. In turn, this increases gradient driver voltage requirements.

Since gradient fields usually have limited FOV, the measurement method should be specified as part of GC gain. Since eddy currents decrease coil gain at high frequencies, a low frequency based measurement method is often used.

GC gain is controlled by the winding geometry, and it can be predicted accurately during design and controlled in production.

4.6.3.2 Field of view

The GC determines gradient subsystem FOV; the other gradient subsystem components have minimal influence on FOV. As distance from the GC center increases, distortion between the actual field and the ideal gradient field increases. This distortion limits the utility of the gradient field for imaging, and becomes.the practical limit on the GC's FOV. Therefore, gradient FOV and GC linearity are intimately related; talking about one without the other is meaningless.

4.6.3.3 Linearity and uniformity

During the gradient coil design cycle, EM modelling tools are used to predict the gradient field linearity.

For the HRMw coils used in Rio, specifications for FOV, Linearity and Uniformity were all defined and used as input parameters during EM design process. The actual measurements for these parameters were close to the design values. (See DOC1739438_HRMW1-1 Test Report, DOC1751524_AH2-2 Gradient Coil Test Results, DOC1816629_HRMW1-002 Test Results for evidence.)

4.6.3.4 Hollow gradient winding conductors

The Rio HRMw gradient coil system is designed and built with hollow; this is the "H" in "HRMw". Hollow wire – insulated copper with a hollow center – is used in all six layers of the gradient coil (x, y, and z windings of both the inner and outer coils).

The hollow conductor tubing provides for coil cooling by allowing liquid to flow internally through the conductors. (Gas could also be used as coolant.) Copper has both high electrical conductivity and high thermal conductivity. The new hollow wire design allowed around 2.5 times better thermal capability over conventional plate coil designs as in XRMB and XRMw coils.

4.6.3.5 Gain vs. bore size tradeoff

The trade-off between coil gain and patient bore diameter is generally considered as one of one of the most important system design decisions. When other factors are held constant, smaller patient bore diameter allows higher gradient capabilities, since smaller bore size allows smaller GC inner diameters. In turn, this allows higher GC gain since a) the distance between primary and shielding layers can increase, and b) the primary layers of the gradient coil are smaller.

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The current market is split into two segments: 60 cm and 70 cm bore sizes. The EM design tools to optimize coil design are key technology for MR system design.

Rio used numerous "scoping studies" to make decisions between gain and bore diameter, along with other parameters. Bore diameter of 60 cm, 65 cm, 68 cm, and 70 cm were discussed. DOC1688918, *Rio Gradient Subsystem Tradeoffs* provides a brief summary. For future design efforts, the whole scoping study results and decision-making process should be captured.

Finalized specifications as a result of year-long discussion can be found in DOC1651768, *HRMw Concept Technical Design (cTDR) Review* package.

4.6.3.6 Gmax and SR

Gmax designates the maximum gradient field strength that the resonant module can support. It is primarily dependent on the maximum current that can be delivered by the gradient amplifier and the gain of the gradient coil. Similarly, SR designates the maximum slew rate that the resonant module can support. It is primarily dependent on maximum voltage that can be delivered by the gradient driver as well as the inductance of the coil. The interaction between the gradient coil and magnet as well as the design of the gradient power supply impacts these values but to a lesser degree.

There are two values that are specified and verified. One value is the clinical value representing the capability of the system in clinical mode and is used as a limit when prescribing and protocol. The second value is used as a datasheet claim and represents the maximum capability of the system design. This capability is usually only achievable under very limited conditions.

There are several methods that have been used to characterize/verify these values. The most precise method is to use a calibrated pickup loop and directly measure the fields generated by the system with eddy current compensation on. A less precise method that has been used in the past is to measure the gradient output current on a scope and convert to field by using the nominal gain of the coil. This method assumes eddy currents are not present (the field profile matches the current profile).

For Rio it was specified that to claim Gmax and SR in the datasheet, the capability had to be able to be used to create an image.

4.6.3.7 Gradwarp

General information on *gradwarp* – geometric distortion correction for gradient non-linearity, applied to reconstructed images – is in section 4.2.1.1.2, *Gradwarp*.

Gradient fields deviate from their ideal field patterns, due to many factors. In addition to physical constraints such as gradient coil length and thickness, the number of windings is limited to keep coil resistance low (minimizing power dissipation) and inductance low (related to attainable slew rate). These gradient field errors result in *image distortions* – incorrect mapping of the object in space due to gradient field non-linearity).

The gradient fields' shape (spatial dependence) can be predicted from each winding's EM design, using Biot-Savart's law. Each axis' shape can be decomposed into a linear combination of spherical harmonic terms; this decomposition is generally performed using a 50cm volume.

Since the Premium Product gradient coils are of a symmetric cylindrical design, only odd order error terms are present. The odd degree terms are used to calculate *gradwarp* coefficients. The *gradwarp* coefficients are stored in gw_coils.dat and used by the *gradwarp* algorithm to correct for gradient field errors (ASL Tech Note 90-70, ASL Tech Note 94-01, ASL Tech Note 99-02).

Prior to the Pioneer product, *gradwarp* corrected up to 5th order odd terms (1st degree); Pioneer extended this capability to 9th order odd terms. The post-*gradwarp* Image distortion due to gradient field non-linearity was analyzed, with results reported in DOC1651768 *HRMw Gradient Coil TDR*. This analysis shows that the reconstructed image pixel position error over a 40cm sphere is

Coil	*X, mm*	*Y, mm*	*Z, mm*	*Correction*
XRMw	1.6	1.6	4.5	5th order
XRMw	0.4	0.4	0.2	9th order
HRMw	0.9	0.9	0.2	9th order

4.6.3.8 Impedance

At low frequencies (below about 200Hz) the GC impedance looks like a simple series inductor-resistor combination, so the impedance looks like:

$$Z = R + jX = R + j\omega L = R + j(2\pi f)L \tag{4.12}$$

In this simple model, the resistance *R* (the real part of the impedance) is constant, while the reactance *X* (the imaginary part of impedance *Z*) increases with frequency.

At higher frequencies, however, this model does not match measured gradient coil impedances. The resistance is not constant, but increases with frequency, and the inductance also changes slightly.

$$Z(f) = R(f) + jX(f) = R(f) + j\omega L(f) = R(f) + j2\pi f L(f) \tag{4.13}$$

An example is useful here. The XRMW coil's x-axis is used; other coils and axes would be similar.

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Fig. 4–20 shows measured impedance data from a XRMW gradient coil's x-axis winding. The change in resistance *R* as frequency increases is remarkable, while the change in inductance (which would cause the *X* data to vary from a straight line) is only seen past ~10KHz.

Resistance *R* is ~0.1Ω at DC, but starts to increase around 100Hz, and reaches ~0.4Ω by 1KHz. At 10KHz, it is ~4Ω, or 40 times the DC resistance.

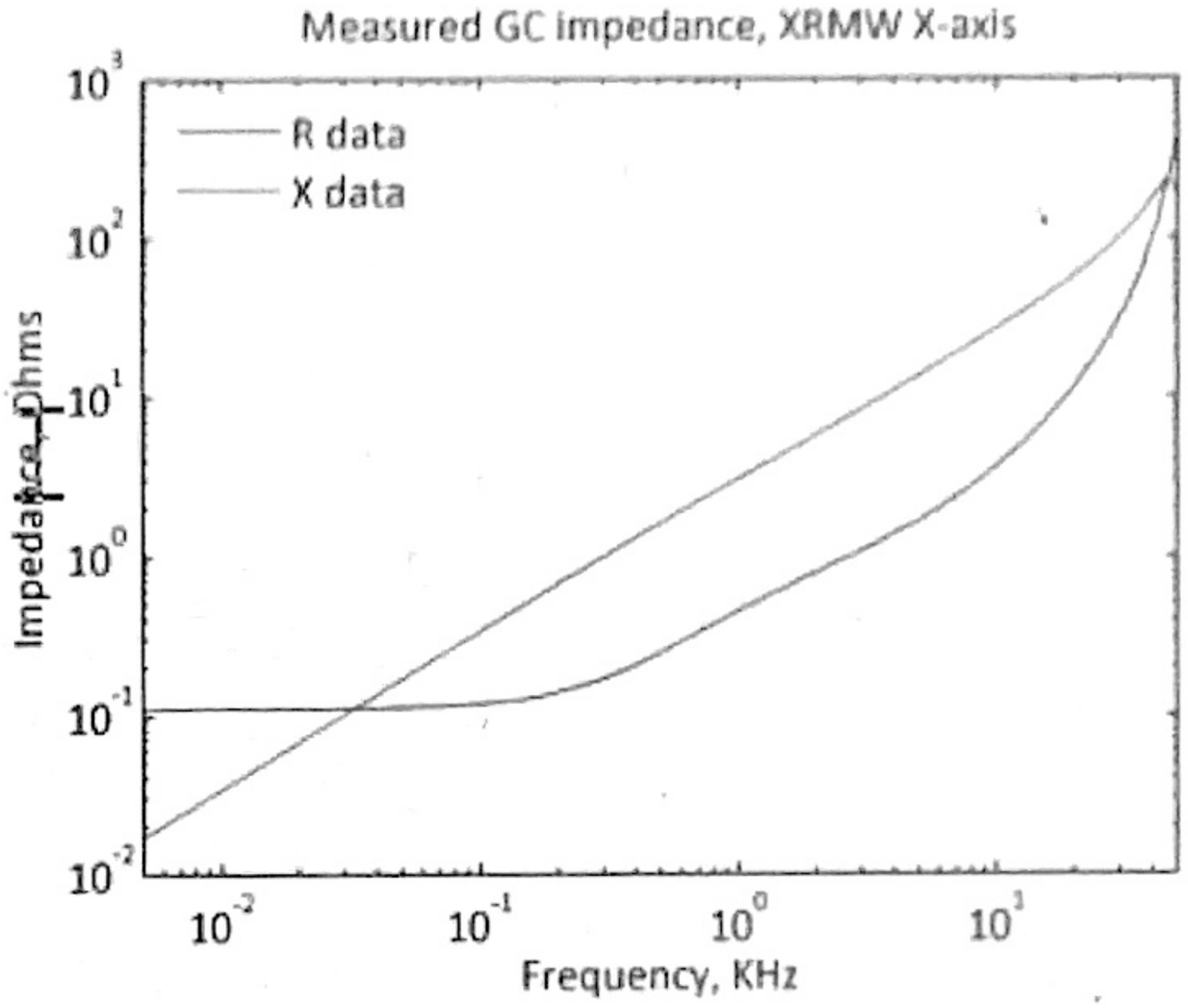


Fig. 4–20: Measured GC impedance

Understanding how the impedance changes is important to understanding several performance aspects: GC power dissipation, temperature of critical components near the gradient coils (the patient bore and passive shim materials, for example), and gradient waveform fidelity (gradient drivers often pre-compensate their output waveforms, based on coil impedance).

4.6.3.8.1 Impedance models

Models for GC impedance are useful, and several methods for modeling how the GC impedance changes with frequency have been developed. One model uses an L-R ladder network, fitted to match measured impedance data. A seven-element model is shown in Fig. 4–21. Once the values for the resistors and inductors are known, the impedance can be calculated.

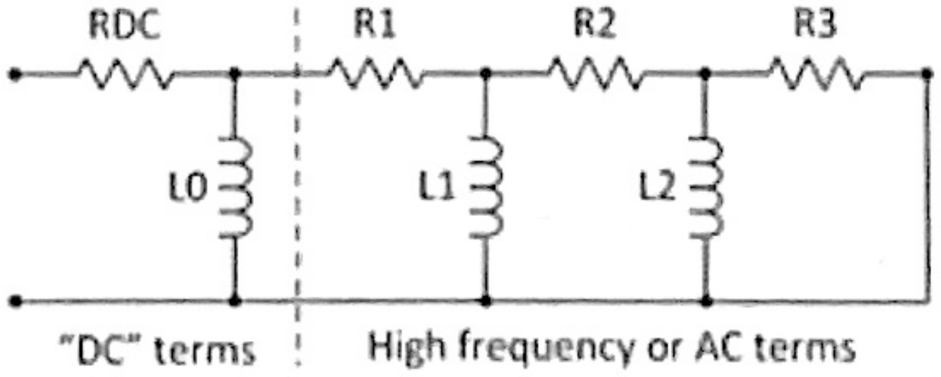


Fig. 4–21: GC impedance model L-R ladder

RDC and *L0* are the low frequency values of resistance and inductance. Additional "AC" terms are added to model the impedance changes. Inereasing the number of elements can extend the either the accuracy of the model, or the frequency range that is accurately modeled.

For the data above (XRMW x-axis impedance), a fitting routine finds the following elements:

RDC=0.110Ω	L0=0.541mH
R1=15.5Ω	L1=2.39mH
R2=75.2Ω	L2=4.07mH

Fig. 4–22 compares the modeled impedance with the measured data. The upper plot compares the impedance curves directly. The lower plot shows the error between the model and the data.

The fitting routine used a 20Hz to 20KHz frequency range, so the model does not capture the upper "tails" on the measured data (which are above 20KHz).

With seven elements, the RMS error is ~1% on resistance *R*, and ~2% on reactance *X*.

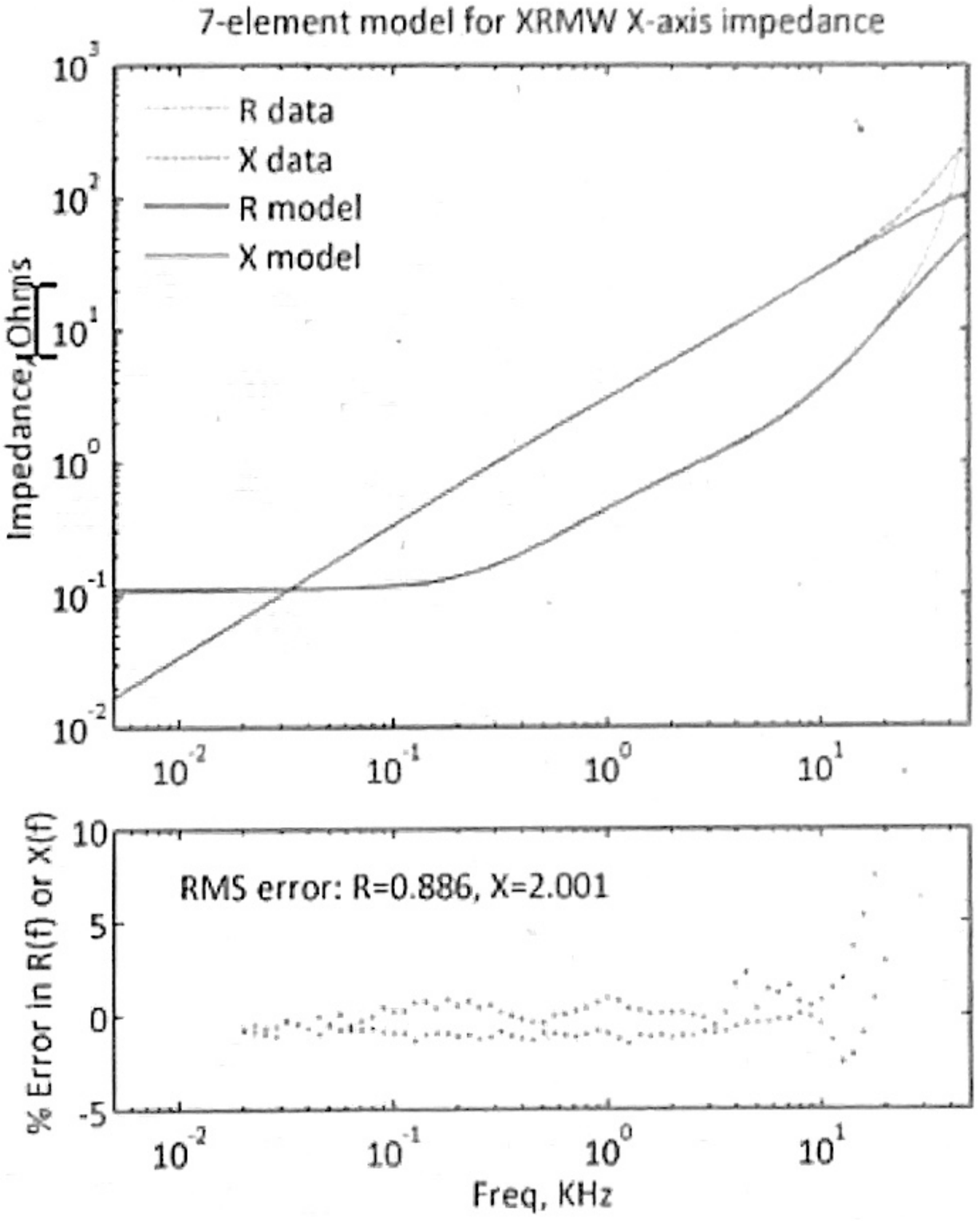


Fig. 4–22: Model for GC impedance

4.6.3.8.2 ABC resistance model

An empirically-derived equation has been developed to characterize *R(f)*, the resistance of the gradient coil as a function of frequency. This is a three-parameter model *A*, *B* and *C*. Only *A*, the DC resistance, has physical meaning.

$$R(f) = A + 10^{-5} B^{*} (\log_{10}|f|)^{C}$$

A pseudocode version makes the operator precedence clear:

```
R(f) = A+1E-5*B*(log10(abs(f)))^C;
```

The frequency-dependent resistance and spectral content of gradient waveforms can be used to calculate the power deposited into the gradient coil. During the scan prescription process, the PSD compares this value to "coilAC_power_rated", "gradient_coil_power_SS_limit_kW" in GradientConfig.cfg to determine if dead time must be added to a sequence.

Method: Coilbuilder calculates the *A*, *B*, and *C* parameters, and analyzes power spectrum following procedures in DOC1448354, *Gradient Coil Heating Calculation*. The coefficients should be reported in the gradient EM test reports and provided to the driver team.

4.6.3.9 EM design considerations

4.6.3.9.1 Self-shielding

Self-shielded gradient coils have two sets of windings: an inner (primary) winding, which creates the main gradient fields, and an outer (shielding) winding, which cancels most of the field in areas outside the gradient coil. This minimizes eddy current response, and minimizes gradient field effects on the magnet.

The annular gap between the inner and outer coils is a stressful environment, for three reasons. First, this area has very high magnetic field intensity, since the inner and outer fields add (as opposed to cancel) in this space. Second, this area is also subject to high physical stress, since the forces on the inner coil tend to be opposite those on the outer coil, which tries to break the material holding the two coils together. Third, temperature rise from gradient coil dissipation provides thermal stress.

4.6.3.9.2 Leakage field

All GE gradient coils use *shielded* coil designs. Shielded coils greatly reduce fields outside the gradient coil, but cannot eliminate them.

Leakage fields are the residual EM fields created *outside* the gradient coil – in regions larger than the OD. Since leakage fields generates numerous problems (eddy currents, magnet heating) GCs are designed to minimize leakage field. For coils of finite length, it is impossible eliminate leakage fields. There are radial and axial components in leakage fields, and radial leakage fields are the main concern in the MR system.

Leakage fields are simulated as part of the EM design process, as this is one of the design constraints. Leakage field simulations accurately predict measured leakage fields; see DOC1751524, *AH2-2 Gradient Coil Test Results* and DOC1816629, *HRMW1-002 Test Results* for evidence.

Test methods can be found in DOC1752400.

4.6.3.9.3 Field roll-off and null point

The *field roll-off point* is the z-position where the gradient of the z-gradient field becomes zero. The roll-off point location is dictated by the placement of gradient windings. It is strongly related to annefact, with a larger roll-off point generally reducing annefact. Increasing the roll-off point comes with the tradeoff of increased number of windings (increased resistance and inductance) or shifting the windings towards the edges of the coil (decreased Gmax, linearity, and uniformity).

The *null point* is the z-position where the x- and y-gradient field becomes zero. Larger null point typically means it can produce larger imaging FOV, however, increasing the null point also means either increasing the number of windings (increasing resistance and inductance) or shifting the windings towards the edges of the coil (decreased Gmax, linearity, and uniformity).

Field roll off and null points can be accurately predicted from EM design simulations. One important constraint is ensuring that Z-roll off point is larger than x/y null points, to avoid aliasing. Test method for field roll-off can be found in DOC1525652.

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4.6.3.9.4 Eddy current

4.6.3.9.5 High order eddy currents

The term HOEC or 'post-compensated eddy currents' refer to the maximum deviation (i.e. error) of the residual time-dependent field to the desired field after eddy current compensation. Eddy current fields impart phase on a spin system, in addition to the desired spatial encoding phase, which can result in ghosting and geometric distortion.

Current spec for Rio's HRMw gradient coil is ±0.44 %; this applies to each axis individually. The predicted performance is 0.26% for the *X*-coil; 0.17% for *Y*; and 0.16% for *Z*, before any contribution from manufacturing variability (DOC1656893).

The test method for HOEC is to the HOEC calibration tool (also called the *grafimage* tool). The test data (rawdata P-files) must be analyzed with a special purpose Matlab tool to get the maximum field error.

4.6.3.9.6 Force, torque, and local pressure

Simulations in the EM design phase evaluate *force, torque,* and *local pressure* to ensure structural integrity of the gradient coil and reduce interactions between gradient coil and other parts of the system.

Force and torque are related to the stability and reliability of the system. Gradient induced mechanical vibration causes 1) high back EMF, 2) narrow resonances in the gradient coil impedance, and 3) disintegration of electrical joints such as jumpers and busbar, which results in arcing and fire.

For symmetrical whole-body gradient coils, torque is typically balanced, but force is not. Since FRMw, the first force balanced gradient coil, all gradient coils are designed to be force balanced. (Ref. DOC1489028 - FRMW3 Net Force Test Results)

When the gradient coil is pulsing, forces created on primary and secondary gradient boards are acting in opposite direction. *Local pressure* is a term that describes the differences in force between primary and secondary coils at given location. Possible failure mode is a internal delamination between gradient layers. In general, higher gradient fields increase local pressure.

4.6.3.9.7 MGI

4.6.3.10 Design Method and Tools

4.6.3.10.1 Roemer 1 code

4.6.3.10.2 Roemer 2 code

4.6.3.10.3 MagStudio

4.6.3.10.4 BEM Code

4.6.3.11 Hollow conductors

4.6.3.12 Mounting and gradient cable connections

4.6.3.13 Power handling

The power dissipated within the gradient coil must be limited for several reasons.

1. To protect the gradient coil itself from damage due to high temperature.
2. To help manage the inner bore temperature, a factor in patient safety.

4.6.3.14 Passive shim compatibility

The XRMw and HRMw have slots in the annular gap between the inner and outer windings, used for passive shims. Due to the demanding environment in the gap, the passive shim materials and parts must have low eddy current susceptibility, high temperature stability, and be tolerant of vibration.

4.6.3.15 RF shield for transmit chain body coil

The XRMB and XRMW both include a conductive RF shield to support the transmit chain's body coil. The shield is a thin mesh, embedded on the inner surface of the primary coil. The shield is highly conductive at RF frequencies (>50MHz, roughly), and so contains the RF transmit field and helps to shield the RF body coil from external noise. At gradient frequencies (up to roughly 100KHz) it has low conductivity, with minimal effect on the gradient field.

The RF shield has a very strong interaction with the RF body coil. The shield's shape (ideally, it would be a perfect cylinder) and alignment (exactly coaxial with the RF body coil) are the main interactions.

4.6.3.16 Temperature sensors

The XRMB and XRMW both include internal thermal sensors, at various locations in the coil assembly. The system uses these sensors to monitor the gradient coil's internal temperature.

Note that prediction and control of gradient coil power dissipation, implemented by pulse sequence calculations, is the *primary* gradient coil temperature control mechanism. The temperature sensors are a *secondary* control mechanism. The controls placed on GC power dissipation should prevent the GC temperature from reaching its temperature limits.

4.6.4 Cabling and filters

Typically, gradient drivers are located in the equipment room; gradient coils are (obviously) in the scan room. Cabling connects the two, and a gradient filter is used to block noise from the scan room.

Gradient cables have to balance current and voltage handling capability against size and weight (which make installation difficult) and cost. Gradient cables also factor into the gradient subsystem capability limits, typically as a limits on per-axis RMS gradient currents.

Gradient filters allow the low frequency (<50KHz) gradient currents to pass, but block both high frequency signals and common mode signals.

4.6.5 AC power distribution

AC power distribution is included in the gradient subsystem design since there are close functional ties. The gradients are the largest power consumers in the system, and the main power distribution components are usually in the same physical cabinet with the gradient drivers.

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4.7 Transmit chain design

The transmit chain (TXC) creates the *B1* RF transmit field, and contains the related components. This includes 1) the exciter, 2) the RF amplifier, 3) RF power monitoring, 4) transmit cables, 5) RF switchgear, 6) the RF whole body coil, and 7) the transmit and coil bias drivers.

Considerations include:

1) Subsystem level design
2) RF frequency, pulse generation, and pulse modulation capabilities
3) RF pulse fidelity
4) RF power capabilities
5) Transmit/receive mode switching capabilities
6) RF power monitoring

4.7.1 Subsystem block diagram

A block diagram for MR750w TXC is shown. (The other systems' diagrams will be similar.)

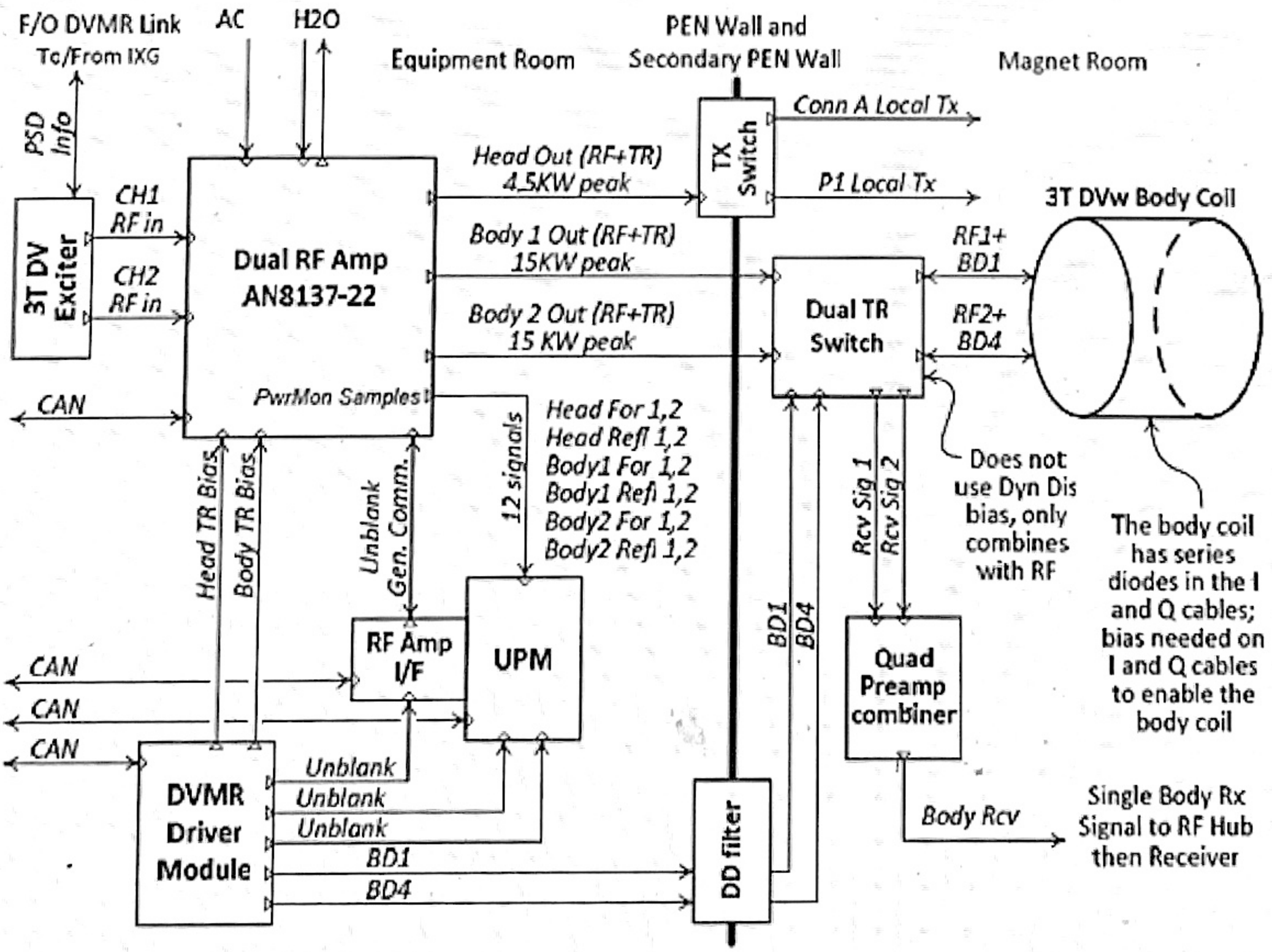


Fig. 4–23: MR750w TXC block diagram

3T DV Exciter

The *3T DV Exciter* creates low power RF transmit pulses for CH1 and CH2 (nominal range: 0 to +4dBm), under system and PSD control. While both channels share a common pulse profile, the exciter can vary the phase and amplitude balance between CH1 and CH2, by adding attenuation and/or phase delay to CH2.

In the 3T dual drive system, the output at exciter J13 leads the signal at J12 by (nominally) 90 degrees.

The Exciter has the capability to independently disable one of the output channels.

The DV Exciter is relatively narrowband with a bandwidth of approximately 650 kHz. If broadband capability is desired in a system, it is necessary to add a different exciter.

The DV Exciter is the source of the Local Oscillator (LO) signal and the system clock signal.

Dual RF Amp

The *Dual RF Amp* (an Analogic Model AN8137-22) amplifies the exciter output, and provides three transmit output ports: BODY_CH1, BODY_CH2, and HEAD. The BODY ports have peak power of 15KW, and the HEAD is 4.5KW. These power levels must be set by calibration, to account for variations in 1) the exciter output (up to 2 dB) and 2) the exciter to amplifier cable attenuation. The amplifiers typically have manual gain (potientiometer) adjustments for full scale power, allowing an input range of -4 to +4 dBm.

Each transmit port has four sampled signals (two forward and two reverse, for a total of 12 signals) for power monitoring, which are used by the UPM (described below).

The amplifier couples the TR bias signals onto the transmit lines.

The amplifier is water cooled.

Other components

The *TX Switch* routes the HEAD transmit signal to either the "A" or "P1" local transmit coil connectors. It is mounted on the pen wall.

The *Dual TR Switch* provides TR switching and preamp protection for BODY_CH1 and BODY_CH2, and couples the disabling bias onto the transmit lines leading the body coil.

The *3T DVw Body Coil* converts transmit power into B1 field, and routes received MR signals back to the body preamps. To minimize site-to-site performance variability, it has field adjustable mechanical centering mounts and tuning adjustment variable capacitors.

The *Quad Preamp Combiner* provides low noise preamplifiers and then combines the CH1 and CH2 receive signals, routing them back to the system receive chain. (The Preamp Combiner and Preamplifiers are part of the receive chain and will be described in more detail in that section.)

The *UPM* (Universal Power Monitor) monitors the sampled signals from the Dual RF Amp, and processes them to calculate the patient SAR. Thru the *RF Amp I/F*, The UPM can blank or disable the Dual RF Amp, and stop the scan, if excess SAR or other fault conditions are detected.

The *DVMR Driver Module* distributes unblank signals, and provides TR bias and body coil disabling bias (BD1 and BD4). All bias lines have real-time monitoring to detect for disconnected loads (open circuit) or excess

bias currents (short circuits). The Driver Module has a total of five dynamic disable driver circuits that have the capability to switch between a positive high voltage (low current) voltage source and a current sink circuit. These provide high voltage to reverse bias PIN diodes or forward current necessary to turn the diodes on. The module also contains three independent TR bias drivers (head, body, MNS) that act as voltage sources and provide the bias to the diodes in the TR Switches in the system.

The *DD Filter* filters the body disable signals as it passes them thru the pen wall.

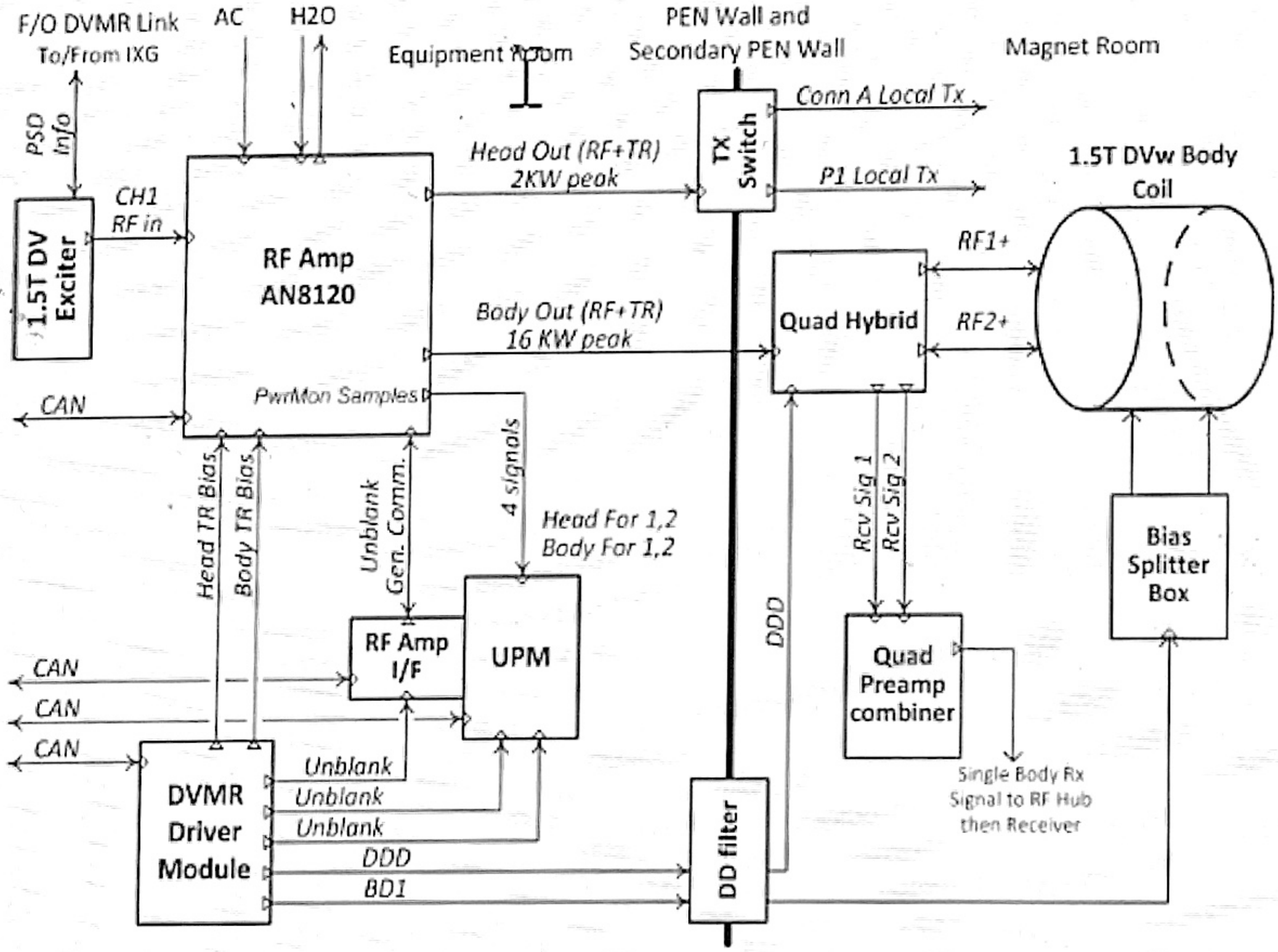


Fig. 4–24: MR450w TXC block diagram

4.7.2 Pulse generation and modulation

The transmit chain's exciter provides the system RF frequency generation, controls the overall RF pulse power level, and provides pulse modulation capabilities.

4.7.2.1 RF frequency generation

The nominal system RF frequency called center frequency, or just *CF*. It is the RF frequency used for both transmit excitation and receive demodulation by default. However, there are many cases where CF is changed or modulated, which will be covered separately.

For many system uses, CF is set to match the MR signal frequency, or Larmor frequency. Given a 1H gyromagnetic ratio of 42.576 MHz/T, nominal CF is approximately 63.864 MHz at 1.5T, and 127.728 MHz at 3.0T.

CF is set by auto prescan, or CF can be set by the user, via manual prescan controls. CF has with a 1Hz resolution.

4.7.2.2 Transmit gain

The transmit gain, or *TG*, control sets the overall power level for a RF pulse. It has 32dB total range, with 0.1dB resolution. TG is often described in "counts"; one count is 0.1dB.

TG is set by auto prescan. Additionally, the top 20dB of TG control range is user-accessible, via manual prescan controls. TG of 200 (counts) generally corresponds to maximum pulse power, and TG 0 is the minimum user-accessible power.

TG is not intended for changing pulse power within a pulse, or changing power between pulses in the same sequence.

4.7.2.3 Pulse modulation

MR imaging and spectroscopy rely on modulated RF pulses. The exciter provides digitally controlled amplitude, phase, and frequency modulation.

Pulse modulation is controlled by the pulse sequence. There are no direct user-accessible modulation controls.

Pulse modulation waveforms can be updated every 2us.

Amplitude modulation is bipolar, with total resolution is 16 bits. This gives a dynamic range of one part in 2^15, or roughly 90dB. Phase modulation uses 16 bits to cover one 360 degree cycle, giving a resolution of ~0.005 degrees per bit.

4.7.3 RF pulse fidelity

RF pulse fidelity ensures that the B1 field created by the TXC matches the intended field, as a function of *time*. (The B1 field's *spatial* dependence is treated separately, under transmit uniformity.)

Typically, pulse fidelity is primarily a function of the *linearity* of the TXC, which describes how flat the gain and phase transfer functions are over the range of output power.

Generally, the exciter, RF amplifier, TR switches, and body coil can all affect pulse fidelity; all use semiconductors, which have the potential to cause non-linearity. In practice, fidelity is limited by the RF amplifier's linearity.

Fidelity specification include both *linearity* and *differential linearity*. Linearity is the deviation from an ideal transfer function; differential linearity is the derivative of linearity, with respect to power.

4.7.4 RF power capabilities

From a system design, adequate RF power is needed to meet the expected B1 capabilities. Since most users are aware that RF power is directly tied to imaging performance, peak RF power is almost always listed on the product data sheet.

Multiple parameters are considered in defining RF power capabilities:

1) Peak power
2) Average power capability
3) Maximum pulse energy
4) Maximum unblank duration

Peak Power:

The peak power capability from the RF amplifier is fairly straight-forward. It is a specified value in the purchase specification and is given as Peak Envelope Power in kW (or dBm) and is guaranteed attainable over a given input power level. It is specified separately for body mode and head mode and in the case of MR750w dual drive, two body output channels are specified. The output power fault trip limit (internal to the amplifier) is set to allow margin above the peak output power to avoid nuisance faults. The peak output power level is set through calibration when the unit is installed in a system and has an input from the exciter at a TG level of 200. All of the amplifiers used in DV are capable of producing the peak power level for pulse widths of 10 msec or less without damage or fault. The peak envelope powers for the various DV systems are shown in the table below:

System and Amplifier	Body Output	Head Output
MR450/MR450w (AN8103 – SRFD2 or AN8120)	16 kW	2 kW
MR750 (AN8135)	35 kW	4.5 kW
MR750w (AN8137)	CH1 = 15 kW; CH2 = 15 kW	4.5 kW

Average Power:

All the amplifiers used in DV have internal fault protection to protect against average power levels that exceed their capability. All amplifiers that use CAN communication (all those except SRFD2) have average power limits in watts as part of the purchase specification for both body and head mode and this average power limit is based on a fixed averaging window length. In addition to this fixed limit, the amplifiers are defined by capability curves that are used by the PSD to determine if the amplifier is capable of the power levels and timing dictated by the sequence. Further information on the amplifier capability curves used by the system can be found in section 4.2.3.2 RF Amplifier Capability (System Safety).

Maximum Pulse Energy:

The capability curve also can be used to define the maximum power (or energy) in a single pulse. Information on the amplifier capability curves used by the system can be found in section 4.2.3.2 RF Amplifier Capability (System Safety).

Maximum Unblank Duration:

In DV systems, the only amplifier that has a hard and fixed limit on maximum unblank duration is the AN8103 or SRFD2 amplifier. For this 1.5T amp, the maximum unblank allowed without fault is a 60% duty cycle. For a given TR cycle, if a sequence requests more than 60% of the time to be in unblank, the amplifier will fault and

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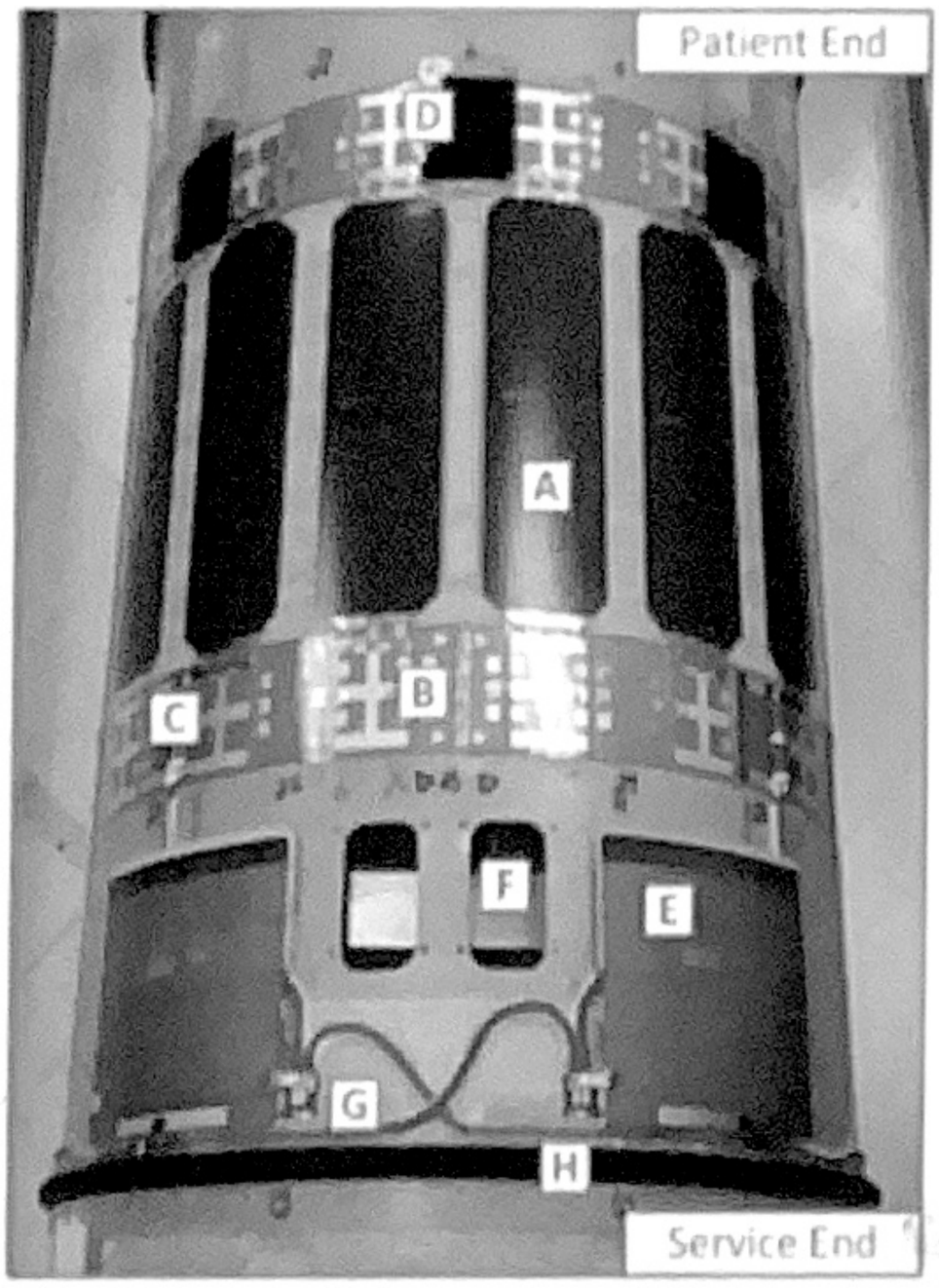
stop the scan. For all other DV amplifiers such as the 3T versions and newer water-cooled AN8120, they can be unblanked indefinitely. In addition there are no restrictions on pulse width; they can operate with a continuous RF output as would be needed for proton decoupling. However, the maximum output power level in this scenario drops significantly from the typical peak output power. For example, the 3T amplifiers can only generate a few hundred watts continuously.

4.7.5 Rio Transmit Coil (Body Coil)

This section covers the design of Rio's body coil.

4.7.5.1 Coil Structure and Basics

The Rio body coil is a transmit/receive, whole body, high-pass 16-run birdcage coil, with two drive inputs, CH1 And CH2. It is similar to other GEMR premium products body coils, but introduces several new features and technology, shown below and described in later sections.



Prototype Rio Body Coil
Viewed from bottom side

A Coil run (16 locations)
B Coil end ring (2 loc's)
C Coil drive point (4 loc's)
D Tuning patch (3 loc's)
E Ground patch (4 loc's)
F Cooling air inlet (2 loc's)
G System drive cabling (2 loc's)
H Cooling air barrier (both ends)

Fig. 4–25: Rio Transmit (Body) Coil

The diagram above shows the Rio body coil, as seen from the bottom view (where most of the significant features are located.)

Additional information about the body coil can be found in:

- DOC1738366, Rio Body Coil TDR.
- DOCxxxxxxx, RF Body Coil Design Guide, Eddy Boskamp, is a great resource for general body coil design and description of important parameters.

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4.7.5.2 Drive Type

The Rio body coil uses a 4-port design, similar to the 750w design. It is a "2x2" 4-port design; there are only two drive inputs from the system (CH1 and CH2), and each input drives two points on the coil. The belief is that 4-port drive improves the body coil's uniformity, although this point is under debate.

Compared to a 2-port design, 4-port designs add complexity and increase the number of cables routed on the coil. An ideal 4-port design requires specific electrical lengths between the drive points. Practical designs may not be able to implement these lengths, and these nonidealities limit the improvement realized with the 4-port design.

Applying decoupling to four points on the coil improves its transparency to the surface coil the body coil is not actively disabled.

One primary factor in choosing a 4-port design for Rio was the planned Digital Surround Technology feature (DST), which uses simultaneous body coil and surface coil receive. True simultaneous acquisition was dropped from the initial Rio release, but the body coil hardware should be compatible if DST is included in a future release.

4.7.5.3 Drive Polarity

The Rio body coil is designed to use CH1 leading, CH2 lagging drive polarity. CH1 is the I-channel, and CH2 is the Q-channel. (This is the convention described in DOC1734279. However, it is different from MR750w, where the BC used CH2 leading polarity.)

The majority of the transmit chain in Rio is reused from the 750w system. However, because Rio uses different drive polarity than MR750w, proper quadrature and elliptical operation required changes to the system exciter. Rio modified the exciter for other reasons, so the polarity related mods were incorporated into these modifications.

DV26 introduced two new MRconfig parameters to describe the body coil: a **bodyCoilType** and **bodyCoilDrivePolarity**. Rio's body coil is type 11, with drive polarity parameter 0; this indicates the exciter outputs should be "unflipped".

By pairing Rio's drive polarity changes and the exciter output configuration changes, the sign of the attenuation and phase values associated with elliptical drive remain the same as 750w. Preset drive mode continues to apply +30 counts and -30 degrees; practically, this attenuates CH2 by 3 dB and moves the two channels farther apart by 30 degrees.

4.7.5.4 Physical Size

While Rio is advertised as a "70 cm" system, the actual mechanical inner diameter of the body coil is 69.5 cm. The primary reason for using 69.5 cm bore diameter is to maximum the body coil to RF shield annular gap; decreasing this gap lowers the body coil efficiency.

The Rio body coil length is the same as 750w; the Kizuna coil was shorter than MR750w. Work to choose body coil length was done early in Rio development. DOC1677082 describes this work, which traded off z-axis uniformity (longer body coil is better) vs annefact (shorter coil is better). The quoted Field of View (FOV) in Rio is 50 cm.

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The body coil is mounted to the magnet/gradient assembly with machined FR4 brackets to minimize acoustic vibration. The brackets provide ±5mm adjustment in both the x and y position.

4.7.5.5 Rung design

The design of the rungs has been optimized thru simulation and testing. Factors considered included E-fields, local SAR, body coil efficiency, and eddy currents (to minimize thermal and acoustic issues).

4.7.5.6 End-Ring Capacitance and Tuning

As a high-pass design, the Rio coil has its resonant capacitors on the end-rings. The design uses both fixed caps and multilayer distributed caps.

Body coil tuning uses the $\sin(\theta)$ method described in 5726301TST, the *Rio Body Coil Test Specification*. The $\sin(\theta)$ method is run on an unloaded body coil, centered in a gradient coil. It uses two tuning probes to measure the body coil's resonant frequency in eight planes (angular directions). This concept for tuning has been used in other DV products, such as 750w and Kizuna.

Tuning has two distinct stages – rough tuning and final tuning. Both must be done inside a gradient coil, because the RF shield is built into the gradient coil. However, rough tuning is done inside a slave gradient coil, while final tuning is done inside the gradient coil that the body coil will be paired with.

1. Rough tuning, which brings the coil close to 127.72MHz, is part of the coil manufacturing process. Rough tuning changes fixed capacitors, soldered at specific locations, and is guided by the $\sin(\theta)$ method. Workers skilled in soldering complete this operation.
2. Once rough tuning is complete, final tuning is done with the body coil installed in the gradient coil it will be paired with, but without the bridge installed. For final tuning, only tuning patches (see below) are used; no soldering should be necessary. The $\sin(\theta)$ method is used to select the proper patch size based on the data from the 8-locations.

Once final tuning is completed, S-parameter measurements are made as a final check of coil tuning and performance. If the coil tuning meets the $\sin(\theta)$ based specifications, the S-parameter performance should also be met.

4.7.5.7 Tuning patches

The Rio body coil introduced "tuning patches", which are assembled into "tuning carriers." A tuning patch is a small circuit board, designed to add capacitance across a Rio body coil end ring capacitor. There are 20 different numbered patches (steps), each providing a different cap value; higher numbers have higher capacitance values. When installed on the coil, each "step" changes the coil's resonant frequency by about 50KHz.

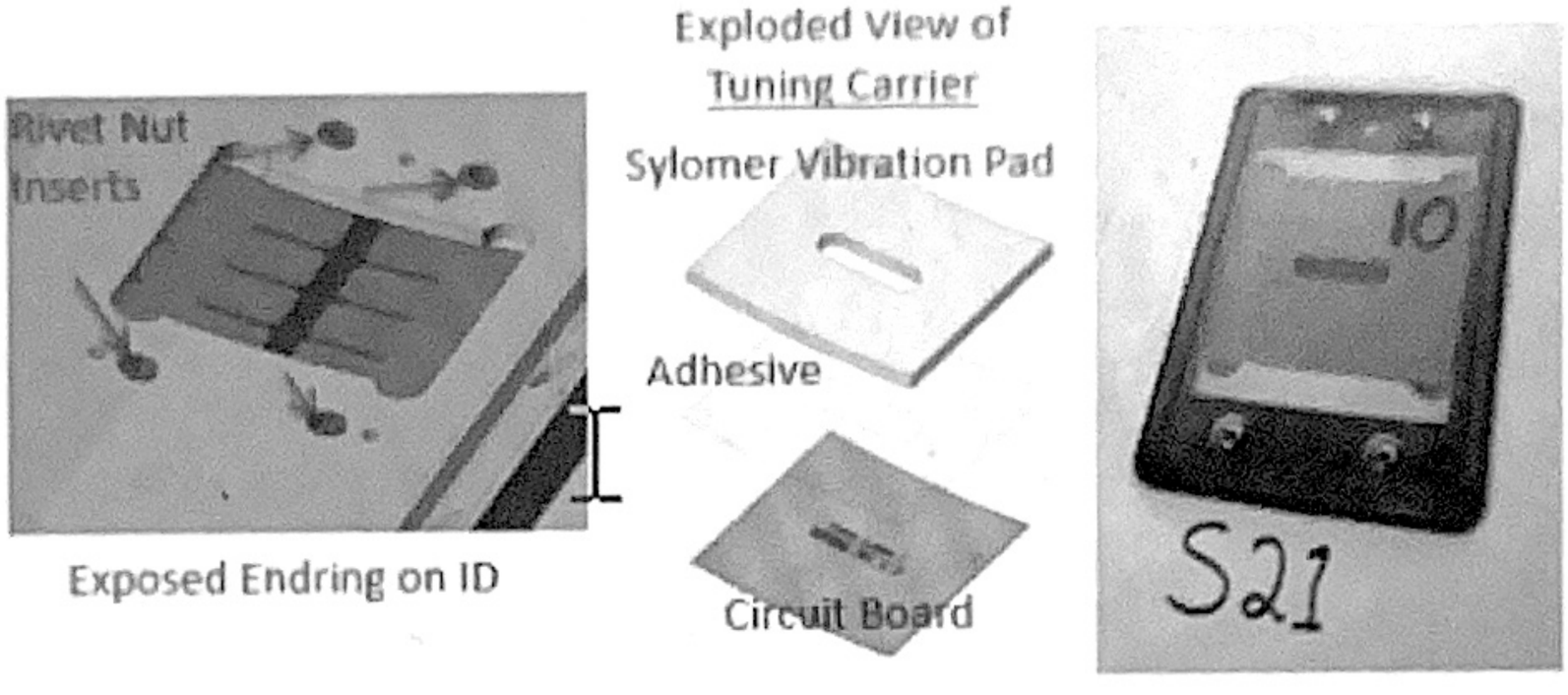


Fig. 4–26: Tuning patches

The Rio coil uses tuning patches in three locations, all on the patient end of the coil. The three patch locations meant to adjust S11 (port 1 impedance), S22 (port 2 impedance) and S21 (isolation). While the adjustments are meant to be independent, the adjustment of one patch can have some impact on the other two parameters.

4.7.5.8 Thermal Event (Arc) Mitigation

Heat spreading material (aluminum nitride, or ALN) is used in 18 locations on the body coil. Most are under fixed tuning capacitors; an arc on or near a fixed cap can create hazardous local heating. The ALN material essentially acts as a heat shield. If an arc occurs during use, the ALN will spread the localized heat, lowering the peak temperature and lowering the risk of a patient bore burn thru.

4.7.5.9 Ground Patch Baluns

The Rio body coil introduced "ground patches" – copper patches that provide a low impedance path (roughly 60Ω) to the RF shield, which is embedded in the gradient coil. The Rio body coil contains four separate ground patches.

The ground patches suppress shield currents on the cables connecting the outside world to the coil's drive points. The patches' low impedance to the RF shield drains excessive shield current to the shield rather than onto the body coil RF cable's outer conductor.

The ground patch serves two functions: 1) it works as a micro-strip bazooka balun by creating a high impedance seen by common mode shield currents at the antenna end of the cable, and 2) it serves to ground the outer shield of the balun to the RF shield of the gradient coil.

The ground patches are copper squares, roughly 15x15cm. Since they are relatively large, they have slits to reduce eddy currents, heating, and acoustic noise. The direction of the slits is important for optimal effectiveness. Because the patches must be close to the RF shield, the coil's acoustic foam (used to dampen vibration and acoustic noise) is not applied to the ground patch.

The distance between the ground patch and the RF cable that is routed below it is critical. This distance affects 1) the body coil's QuadCal attenuation contribution, and 2) coil reliability due to heat generation under the ground patch (charring and bore wall heating). The distance is controlled by design and manufacturing controls.

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4.7.5.10 Acoustic treatment

The body coil has many conductors and conductive patches. Gradient fields induce currents in these conductive materials, causing vibration and acoustic noise. The body coil has acoustic foam strategically applied to minimize the coil's acoustic noise contribution.

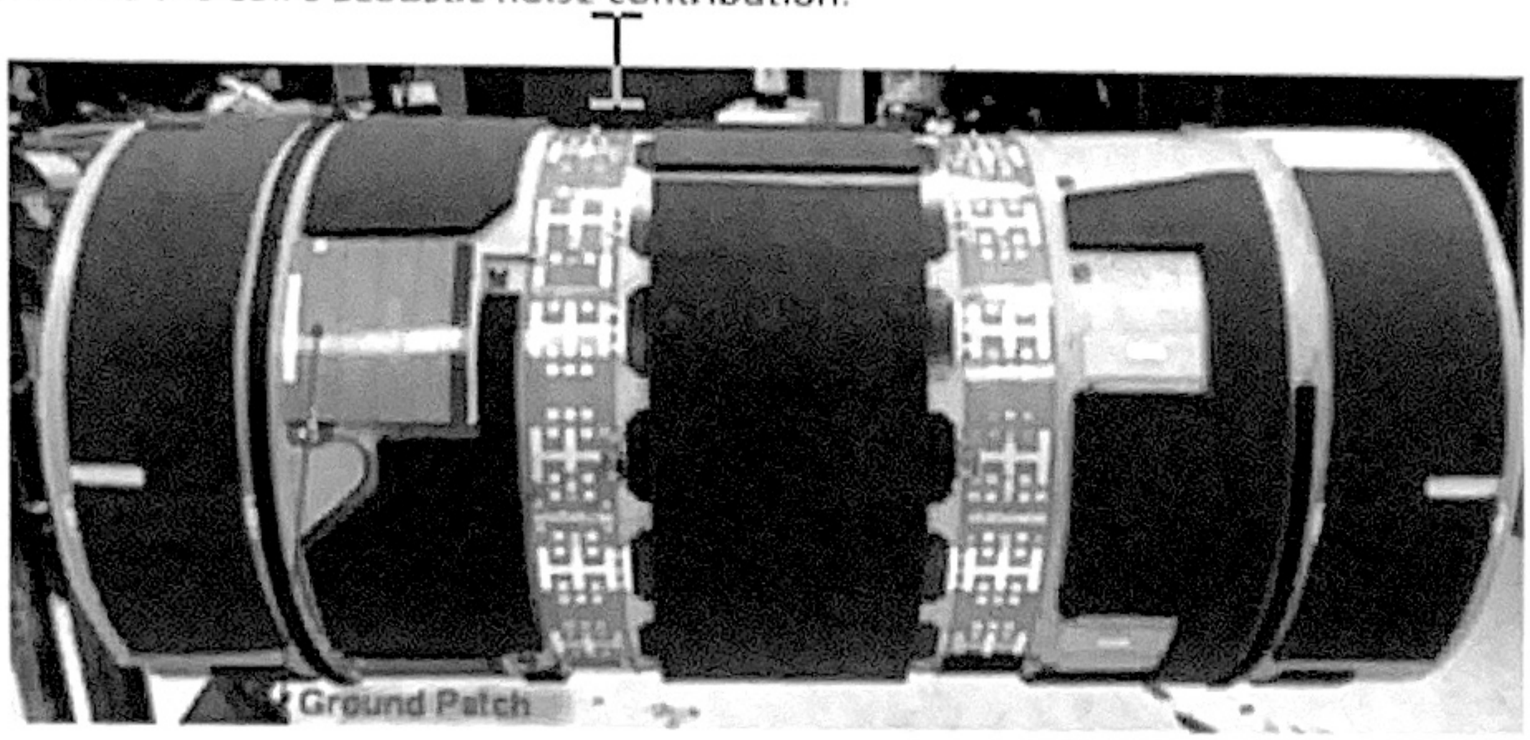


Fig. 4–27: Acoustic Foam

4.7.5.11 Remote Decoupling

PIN diodes are used to enable/disable the Rio body coil. The diodes are back biased with +1000V to enable, and carry bias current to disable the coil. The Driver Module supplys the bias signal; only one driver is used, the DD output.

The decoupling circuits are in a separate module, the DD Box, that is remote from the body coil. With a separate module, a PIN diode failure will not require replacing the body coil assemply; only the DD Box needs to be replaced. This is a service advantage, and saves body coil tuning on-site because the coil component is not touched.

The PIN diode's impedance is transformed via a multiple of a half-wavelength to the appropriate point on the body coil. The electrical phase length between the coil drive points and the decoupling circuits is critical, and must be phase length controlled.

The DD Box is aluminum to provide solid RF grounding for the circuit board. To improve Rio acoustic performance, a "keep-out" zone was defined for aluminum structures; this zone is approximately 2 feet around the circumference of the bore. To comply, the DD Box is mounted with a metal bracket, attached to magnef ground, on the lower rear face of the magnet, near the edge.

The RF power cables connecting the DD Box to the DTRSW include bazooka baluns to suppress shield currents and reduce interference with other electronics on the side of the magnet.

Disabling in a separate module is is unique to Rio. (The 750w body coil has a form of remote disabling. The diodes were physically in the coil assembly, and internal controlled length cables connected the diodes to the coil's resonant structure.)

4.7.5.12 2-channel receive

A new Rio body coil feature (although it does not directly impact the RBC design) is the use of two body coil receive channels, which are separate up to the reconstruction subsystem, where recon software combines them in quadrature. Other premium systems use a *hardware* quadrature combiner, located after the preamps; the combined signal is routed to recon.

4.7.6 T/R mode switching

Almost all MR imaging separates RF transmit pulses and MR signal reception into separate time intervals. The system has a two-state *TR mode*, with the states predictably named XMIT and RCV. The TR mode toggles with a group of control signals, collectively called "transmit unblank" or just "unblank". When the system is "unblanked", it is in XMIT mode; if it is "not unblanked", it is in RCV mode.

In addition to these *transmit* unblank signals, there is also 1) a signal to blank and unblank the exciter, and 2) a signal to blank and unblank the receivers.

4.7.6.1 Transmit Unblank

In DV systems, three transmit unblank signals are used, called NB (narrowband), BB (broadband) and CW (continuous wave). These signals are controlled by the PSD, created in the exciter, and used by the transmit chain. Originally, the design intent was to have separate, independent control of three separate RF amplifiers that could be used:

1. The narrowband signal (NB UNBLK) controls the "main" RF amplifier, which is used for almost all imaging and most 1H spectroscopy. When someone refers to "the RF amp", they are almost always referring to the NB amplifier.
2. The broadband signal (BB UNBLK) control a separate, stand-alone amplifier used for Multi-Nuclear Spectroscopy (MNS). MNS is not covered in this section.
3. The continuous wave (CW UNBLK) signal was to be used continuous wave RF amplifier. This amp was planned to support "decoupling" – a special imaging option used with spectroscopy. However, this was never put into product, and a CW amplifier has never been part of the DV system.

In addition to driving the RF amplifiers, each unblank signal is provided to the UPM (which is used for RF power monitoring). If the UPM detects any RF power outside of the corresponding unblank window, a fault is generated.

4.7.6.1.1 Dual Drive and CW UNBLK

MR750w's introduction of Dual Drive required two separate body transmit channels. The MR750 "BODY" transmit channel was replaced in MR750w by "BODY CH1" and "BODY CH2". Since the RF power from *all* transmit channels need to be monitored, the allocation of transmit channels to the UPM needed to change. Each UPM has four "blocks" of detectors. In MR750, the CW amplifier was never implemented but was assigned to UPM block 4; therefore, UPM block 4 was reallocated to monitor MR750w's BODY CH2.

The tables below compares the transmit channels, unblank signals, and UPM allocations for MR750 and MR750w.